

Experiment 1

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Challenges

- ChatGPT Timeout error
- ChatGPT Rate limit error
- JSONDecodeError
- Program running slowly

ChatGPT Timeout error

Reason: This error is a ChatGPT internal error.

Solution: Try again a few moments later.

Retries = 3

While retries >0:

Try call chatgpt

Except Timeout:

Retries = retries -1

Sleep 60 seconds

ChatGPT rate limit error

Reason: Quota is exceeded.

Solution: We should charge API key again.

Price: 0.002\$ / 1000 tokens

2703 transcriptions, each transcription has approx 200 tokens,

For each experiment $2703 \times 200 \times 0.002 / 1000$ \$ is needed.

JSONDecodeError

- Reason: ChatGPT does not respond in the correct Json format for examples including quotations
- Possible solution: 1) improving prompt 2) Using GPT-4

Example:

transcription: Don't cry, don't cry. He whispered.

ChatGPT raw output: {"text": "Don't cry, don't cry," he whispered."}

Program too slow

Reason: Because of sending too many request over internet to ChatGPT and waiting for response.

Solution: I used multi threading which is suitable for sending request.

Prompt / query

```
messages = [
```

```
    {'role': 'system', 'content': f""""You are an assisting AI specialized in correcting ASR errors. You will be presented with an ASR transcription delimited by {delimiter} characters, and your task is to rectify any errors in it. Please provide your output in JSON format with the key "text." The text within "text" should be enclosed in double quotation marks. If the text contains any internal quotations, escape them with a backslash (\). Do not output any additional text that is not in JSON format. Do not write any explanatory text after outputting the requested JSON. """"},
```

```
    {'role': 'user', 'content': f'{delimiter}the day he is coming, said Pomeethias. When Jupiter will send a flood the destroy mankind from the earth.{delimiter}'},
```

```
    {'role': 'assistant', 'content': '{"text": "the day is coming said prometheus when jupiter will send a flood to destroy mankind from the earth."}'},
```

```
    {'role': 'user', 'content': f'{delimiter}{asr_transcription}{delimiter}'
```

```
]
```

Results

WER_original : 8.5339%

WER_corrected_chatgpt : 8.4123%

CER_original : 3.5397%

CER_corrected_chatgpt : 4.3189%

SER_original : 61.1765%

SER_corrected_chatgpt: 62.1252%

My team's suggestions about experiment-1

Enno suggested me to get response from ChatGPT in text format instead of json format to avoiding JSONDecodeError.

He also suggested me to start working with confidence score in sentence level.